Microeconomics I

Class 6 — Expenditure minimisation and the cost of living

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Slide 1 Title

Class 6 reads class 4's problem backwards: not the best bundle a budget can buy, but the cheapest bundle that reaches a given utility level. The answer brings two new functions, the expenditure function and compensated demand, one lemma, and the right way to measure a change in the cost of living, which class 1 promised for this session. MWG 3.E–3.G.

Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Quiz Start of class, on paper

A short quiz on paper opened the class: circle the letters of your answers in the block on the sheet, write your name on it, and hand in only the strip. Problem set 1 was collected in the same round, one copy per group.

Plan for today

10:00	Short quiz
10:10	Warm-up; two price indices
10:20	Expenditure minimisation: *h* and *e*
10:31	In pairs: kinks and corners
10:38	Relabelling; properties of *e* and *h*
10:50	Break
11:00	Shephard's lemma; duality
11:11	Compensated price effects; net substitutes
11:19	The true cost of living
11:25	In pairs: fixed basket vs true index; summary

Slide 2 Plan for today

Before the break: the expenditure minimisation problem, its value $e(p, u)$ and its solution $h(p, u)$, first for $x_1 x_2$ and then for maps with kinks and corners; the identities that tie e to class 5's v; and what every e and every h must satisfy. After the break: Shephard's lemma, the price picture that makes e concave, compensated price effects, and the payoff, the true cost-of-living index against a fixed basket.

MWG 3.E–3.G; Muñoz-Garcia chapter 2 is the gentler read.

Practicalities

- Problem set 1: on the desk at the start of class, one paper copy per group
- Problem set 2: handed out in class today, one per group;
  due at the start of class, Wed 21 Oct
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

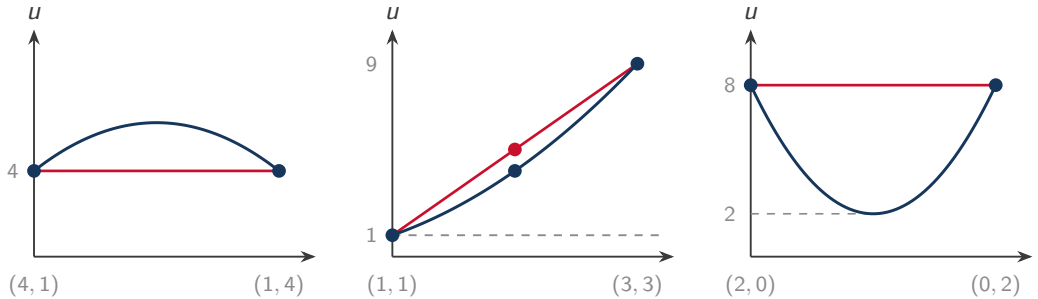
Slide 3 Practicalities

Problem set 2 is handed out on paper at the end of today's class, one sheet per group, each group its own sheet; it is not posted online. It covers classes 4 to 6, utility maximisation, indirect utility and today's expenditure minimisation, and is due at the start of class on Wednesday 21 October, on paper, one copy per group, the answer sheet with all names on top of the working.

Office hours are on Wednesdays, 14:00–15:00, Room 445, up to 20 minutes per student; booking by e-mail gives priority. Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu.

Warm-up: concave, or only quasiconcave?

Utility along the segment between two bundles. What does each picture rule out?



x_1x_2 **from** $(4,1)$ **to** $(1,4)$ — nothing ruled out

x_1x_2 **from** $(1,1)$ **to** $(3,3)$ — not concave

$x_1^3 + x_2^3$ **from** $(2,0)$ **to** $(0,2)$ — not quasiconcave

Concave: never below the chord. Quasiconcave: never below the lower end.

Class 3, slides 21–22; MWG 3.C, pp. 49–50.

Two US price indices, the same prices: which rises faster?

United States: two official price indices, the same prices

CPI-U: today's prices of an earlier basket (Laspeyres)

Chained CPI-U, since 2002: the basket follows what people buy

Which rises faster, and by how much a year?

0.02 · 0.2 · 2 percentage points (average, 2001–2023)

U.S. Bureau of Labor Statistics: CPI-U and Chained CPI-U documentation.

Slide 4 Warm-up: concave, or only quasiconcave?

Each panel is a utility function restricted to the segment between two bundles, class 3's test. Concave means that at every mix of the two bundles utility is at least the same mix of their utilities, so the graph stays on or above the chord; quasiconcave means utility is at least the smaller of the two, so the graph stays on or above the lower end point, which is the same as convex upper contour sets, that is convex preferences.

Panel A sits above both, so it refutes nothing, but one segment cannot prove either property. Panel B is the same function x_1x_2 on the diagonal, where it rises like t^2: the midpoint $(2,2)$ gives 4 against the chord's 5, so x_1x_2 is not concave, though it never drops below 1 and is in fact quasiconcave. Panel C, $x_1^3 + x_2^3$, gives 2 at the mix $(1,1)$, less than both ends: not quasiconcave, the bowed-out curves of class 3.

Why today: the cheapest way to a curve is unique under strict quasiconcavity (slide 14), never needing concavity, while today's expenditure function is concave in prices for every consumer, even C (slides 13 and 19). The trap is to read convex indifference curves as concavity of u: they only say quasiconcave.

Slide 5 Two US price indices, the same prices: which rises faster?

The United States publishes two price indices from the same price quotations; they differ only in the basket. The CPI-U prices a basket fixed in an earlier period, a Laspeyres index; the chained CPI-U, published since 2002, lets the basket follow what households actually buy from month to month.

Class 1's slide 6 reported that the old US index overstated the rise in the cost of living by about 1.1 points a year, 0.4 of it because people substitute towards whatever has become relatively cheap, and promised the fix for this session: the chained index is the statistical office's answer to that part. Which index rises faster, and by how much, is the next slide; the trap is to think an index is just an average of price changes, so that the basket cannot matter.

Source: the U.S. Bureau of Labor Statistics' documentation of the CPI-U and the chained CPI-U.

The earlier basket runs ahead: about 0.2 points a year

CPI-U, earlier basket

+0.2

points a year above the chained CPI-U (average, Dec. to Dec., 2001–2023)

since 2017

tax brackets

US federal brackets rise with the chained CPI-U

Why a fixed basket runs ahead of the true cost of living: after the break.

BLS, Chained CPI: questions and answers (bls.gov/cpi); BLS, Introducing the Chained CPI.

Slide 6 The earlier basket runs ahead: about 0.2 points a year

From 2001 to 2023 the December-to-December rise of the CPI-U exceeded that of the chained CPI-U by about 0.2 percentage points a year on average, varying from year to year: the earlier basket runs ahead. The chained index weights categories by current spending (a Törnqvist formula), so households' substitution between categories counts; the CPI-U holds the earlier basket. The gap is of the same kind as the between-category part of class 1's substitution bias, which the Boskin Commission had put at 0.15 of its 0.4 points.

Since 2017 US federal tax brackets are indexed to the chained CPI-U, so they rise a little more slowly than they would with the CPI-U. Why a fixed basket must overstate the true cost of living, and by how much for one consumer, is the second half of today; the chained index is closer to the truth, not equal to it.

Expenditure minimisation

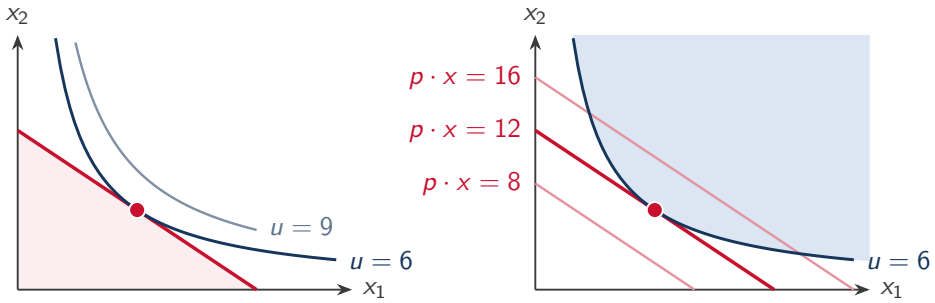
MWG 3.E

Slide 7 Expenditure minimisation

Part 1: the expenditure minimisation problem, class 4's utility maximisation read backwards. MWG 3.E.

Two problems, one tangency: fix the budget, or fix the curve

EMP: $\min_{x \geq 0} \; p \cdot x$ s.t. $u(x) \geq u$ value $e(p,u)$, solution $h(p,u)$



UMP: $w = 12$ fixed
the highest curve the line reaches: $u = 6$

EMP: $u = 6$ fixed
the lowest line that reaches the curve: 12

Same bundle $(3,2)$: the best bundle 12 euros buy is the cheapest way to reach $u = 6$.

MWG 3.E, p. 58: Figure 3.E.1, Proposition 3.E.1; $u = x_1 x_2$, $p = (2,3)$.

For $x_1 x_2$: compensated demand and the least cost of u

tangency $x_2/x_1 = p_1/p_2$, on the curve $x_1 x_2 = u$:

$$h(p,u) = \left(\sqrt{u\, p_2/p_1},\ \sqrt{u\, p_1/p_2} \right)$$

$$e(p,u) = p \cdot h(p,u) = 2\sqrt{u\, p_1 p_2}$$

at $p = (2,3)$, $u = 6$: $h = (3,2)$, $e = 12$

No w anywhere: h and e depend on prices and the target u.

MWG 3.E, pp. 59–61; Example 3.E.1, p. 63, does $x_1^\alpha x_2^{1-\alpha}$; ours is $x_1 x_2$.

Slide 8 Two problems, one tangency: fix the budget, or fix the curve

The two problems share one picture and one tangency but hold different things fixed. The utility maximisation problem holds the budget line and looks for the highest curve it touches; the expenditure minimisation problem holds a curve, the utility level to be reached, and lowers the cost line until it only just touches. Its value is the expenditure function $e(p,u)$, a number of euros, and its solution is the Hicksian or compensated demand $h(p,u)$, a bundle; the constraint is on utility, not on money.

Proposition 3.E.1 says the two problems pick the same bundle: with continuous, locally non-satiated preferences, a bundle that is best at wealth w is also the cheapest way to its own utility level, and the minimum cost is w, and conversely. Local non-satiation is what makes it work: if some bundle reached $u = 6$ for less than 12, local non-satiation would give a better bundle next to it that still costs under 12, so $(3,2)$ would not have been the best that 12 euros buy (class 2). That is why $(3,2)$ answers both questions.

Slide 9 For $x_1 x_2$: compensated demand and the least cost of u

The same two steps as class 4's tangency, with the roles swapped: the tangency $x_2/x_1 = p_1/p_2$ is kept, but instead of lying on the budget line the bundle must lie on the curve $x_1 x_2 = u$. Solving gives h, and pricing h gives e; the square roots make visible that each compensated demand falls in its own price and rises in the other.

Note what is absent: wealth. h and e take prices and a target utility, nothing else; and h_1 depends on p_2 although class 4's Walrasian $x_1 = w/(2p_1)$ does not, which slide 21 explains. MWG's Example 3.E.1 does the general $x_1^\alpha x_2^{1-\alpha}$; the first-order conditions are class 4's with the roles of p and ∇u swapped.

e and v undo each other: four identities

identity	check for $x_1 x_2$
$e(p, v(p,w)) = w$	$2\sqrt{\dfrac{w^2}{4p_1p_2}\,p_1p_2} = w$
$v(p, e(p,u)) = u$	$\dfrac{e(p,u)^2}{4p_1p_2} = u$
$h(p,u) = x(p, e(p,u))$	$\dfrac{e(p,u)}{2p_1} = \sqrt{u\,p_2/p_1}$
$x(p,w) = h(p, v(p,w))$	$\sqrt{v(p,w)\,p_2/p_1} = \dfrac{w}{2p_1}$

Know v? Invert it in its last argument: that is e.

MWG Prop. 3.E.1, p. 58; (3.E.1), p. 60; (3.E.4), p. 62. $v = w^2/(4p_1p_2)$: class 5, slide 8.

Slide 10 e and v undo each other: four identities

Rows 1 and 2 say that at fixed prices $e(p,\cdot)$ and $v(p,\cdot)$ undo each other: the best curve that w reaches costs exactly w to reach, and spending $e(p,u)$ well lands exactly on u. So e is the inverse function of v in its last argument, not a reciprocal: to get e from a known v, solve $v(p,w) = u$ for w.

Rows 3 and 4 explain the name compensated: $h(p,u)$ is what the consumer would buy if her wealth were adjusted to exactly $e(p,u)$, enough to keep her on u as prices move (MWG's Hicksian wealth compensation, Figure 3.E.4), and Walrasian demand is compensated demand at the utility that w buys. The checks substitute class 5's $v = w^2/(4p_1p_2)$ and class 4's $x_1 = w/(2p_1)$ into slide 9's formulas; in row 1, p_1p_2 cancels under the root and $2\sqrt{w^2/4} = w$ is left.

In pairs: the cheapest way to the curve, with kinks and corners

$p = (2,3)$: the cheapest bundle that reaches u; then $e(p,u)$ for any p.

$u(x)$	reach	cheapest bundle	$e(p,u)$	
$\min\{x_1, x_2\}$	$u = 2.4$	$(2.4,\ 2.4)$	$u\,(p_1 + p_2)$	the kink: she buys pairs
$x_1 + x_2$	$u = 6$	$(6,\ 0)$	$u \min\{p_1, p_2\}$	all on the cheaper good
$x_1^3 + x_2^3$	$u = 216$	$(6,\ 0)$	$u^{1/3} \min\{p_1, p_2\}$	a corner; the tangency is the dearest point

Four minutes.

Every row costs 12: $e(p, v(p,12)) = 12$ again.

Slide 11 In pairs: the cheapest way to the curve, with kinks and corners

The three utility levels are what $w = 12$ buys at $p = (2,3)$: class 5's warm-up consumers and perfect substitutes. None has a tangency to use, so the recipe is the picture: draw the curve, lower a cost line until it just touches.

Row 1: the curve is an L with its corner at $(2.4, 2.4)$, and any downward-sloping cost line first touches it there; she buys the goods in pairs and a pair costs $p_1 + p_2$, so $e = u(p_1 + p_2)$, class 5's v solved for w. Row 2: the curve is a straight line, so the cheapest point is the end on the cheaper good; at $p_1 = p_2$ every point is cheapest and h is a set. Row 3: the curve bows outwards (the warm-up's panel C), so the point where a cost line is tangent, about $(4.24, 5.19)$ at a cost of about 24, is the dearest point on the curve and the cheapest is the end $(6,0)$; applying MRS $= p_1/p_2$ there is the trap.

Rows 2 and 3 are different preferences with the same e up to the curve's label: the expenditure function cannot see whether the upper contour set is convex. And each row costs 12, which is row 1 of slide 10 again.

Relabel the curves: the cheapest bundle and its cost stay put

u	$v(p,w)$	$e(p,u)$	the curve through $(3,2)$
x_1x_2	$\dfrac{w^2}{4p_1p_2}$	$2\sqrt{u\,p_1p_2}$	$u=6$: $e=12$
$\frac{1}{2}\ln x_1+\frac{1}{2}\ln x_2$	$\ln w-\frac{1}{2}\ln(p_1p_2)-\ln 2$	$2\exp(u)\sqrt{p_1p_2}$	$u=\frac{1}{2}\ln 6$: $e=12$

Replace u by $f(u)$: v becomes $f(v)$; e and h keep euros and bundles, only the label changes.

MWG p. 57 ($\tilde{v}=f(v)$) and (3.E.1), p. 60; class 5, slide 12.

Slide 12 Relabel the curves: the cheapest bundle and its cost stay put

Class 5's pair again: $\frac{1}{2}\ln x_1+\frac{1}{2}\ln x_2$ is $f(x_1x_2)$ with $f(t)=\frac{1}{2}\ln t$, the same preferences, and class 5 showed that v becomes $f(v)$. Inverting the second v in w gives the second e. The two formulas look different only because they use different labels for the same curve, 6 or $\frac{1}{2}\ln 6$; reaching the curve through $(3,2)$ costs 12 either way, at the same bundle.

The expenditure problem only sees the curve, never its number: in symbols $\tilde{e}(p,f(u))=e(p,u)$ and $\tilde{h}(p,f(u))=h(p,u)$. So e's euros and h are properties of the preference, while v and class 4's λ are properties of the representation, and that is why e, a sum of euros, is the right yardstick for welfare on slide 22. One more contrast: both e's are concave in p, but in u the first is concave and the second convex, so concavity in u is about labels (the warm-up) and concavity in p is not.

$e(p,u)$: homogeneous of degree 1 in p, increasing in u, concave in p

property	why	for $e=2\sqrt{u\,p_1p_2}$
(i) homogeneous of degree 1 in p	same curve, every cost $\times\alpha$	$2\sqrt{u\,\alpha p_1\,\alpha p_2}=\alpha\,e$
(ii) strictly increasing in u	continuity of u	$\dfrac{\partial e}{\partial u}=\sqrt{\dfrac{p_1p_2}{u}}>0$
nondecreasing in each p_ℓ	no bundle gets cheaper	$\dfrac{\partial e}{\partial p_1}=\sqrt{\dfrac{u\,p_2}{p_1}}>0$
(iii) concave in p	after the break	
(iv) continuous		

Tuesday's v: degree 0 and quasiconvex. Today's e: degree 1 and concave.

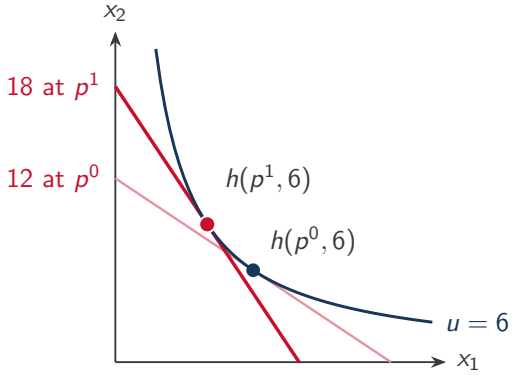
MWG Proposition 3.E.2, p. 59: u continuous, $\succsim$ locally non-satiated, $X=\mathbb{R}^2_+$.

Slide 13 $e(p,u)$: homogeneous of degree 1 in p, increasing in u, concave in p

Proposition 3.E.2 lists what every expenditure function satisfies under a continuous u and local non-satiation, and each line has a one-line reason. Degree 1 in prices: doubling all prices leaves the curve where it is, so the same bundle stays cheapest and its cost doubles; contrast v, where doubling prices and wealth together changes nothing. Strictly increasing in u: if a higher curve cost no more, scale its bundle down a little; by continuity it still beats the lower level and costs less.

Only nondecreasing in each price: a dearer good cannot make reaching u cheaper, but if she buys none of it its price does not matter, as in row 2 of Exercise 1, where p_2 drops out of $6\min\{2,p_2\}$ while $p_2>2$. Concavity in p is the picture after the break, and it needs no convexity of preferences: row 3 of Exercise 1 has bowed-out curves and an e that is still concave in p. Keep the third column's $\partial e/\partial p_1$ in mind: it is h_1 from slide 9.

$h(p, u)$ stays on the curve and never rises with its own price



(i) $h(\alpha p, u) = h(p, u)$

only relative prices matter

(ii) $u\big(h(p, u)\big) = u$

no utility to spare

(iii) $\succsim$ strictly convex: one bundle

convex: a convex set of bundles

$(p^1 - p^0) \cdot \left(h^1 - h^0\right) \leq 0$

here $(2.5,\ 0) \cdot (-1,\ 1) = -2.5$

Good 1 dearer, utility held at $u = 6$: she buys less of it here, and never more in general.

MWG Prop. 3.E.3, p. 61; Prop. 3.E.4 and (3.E.5), p. 62; $u = x_1 x_2$, $p^0 = (2, 3)$, $p^1 = (4.5, 3)$.

Slide 14 $h(p, u)$ stays on the curve and never rises with its own price

Proposition 3.E.3, for compensated demand. Degree 0 in prices, because only the slope of the cost line matters: $p^1 = (4.5, 3)$ is 1.5 times $(3, 2)$, so the cheapest bundle at p^1 is the cheapest bundle at $(3, 2)$, which is class 5's slide 10 bundle $(2, 3)$. No utility to spare: a bundle above the curve could be shrunk a little, still reach u, and cost less. Uniqueness needs strictly convex preferences, which is strictly quasiconcave u in the warm-up's words, never concavity; with flat pieces (Exercise 1, row 2 at $p_1 = p_2$) the cheapest set is a segment.

The last line is the compensated law of demand. Each bundle is cheapest at its own prices, so $p^1 \cdot h^1 \leq p^1 \cdot h^0$ and $p^0 \cdot h^0 \leq p^0 \cdot h^1$; subtracting gives the inequality, and with one price moving it says compensated demand never rises in its own price. Walrasian demand has no such law, class 5's rice was Giffen; so when ordinary demand slopes up, the culprit is the wealth effect of the price change (her purchasing power moves), not substitution.

(optional) Both prices move: which bundles can be the new h?

Utility target u fixed. At $p^0 = (2, 3)$ her cheapest bundle is $h^0 = (3, 2)$.
Prices move to $p^1 = (3, 1)$. Could $h(p^1, u)$ be this bundle?
Check slide 14's law and the definition of h, at both price vectors.

(1) $(2,\ 4)$ yes: law $-5 \leq 0$; costs $16 \geq 12$ at p^0, $10 \leq p^1 \cdot h^0 = 11$ at p^1

(2) $(4,\ 1)$ no: law $(1, -2) \cdot (1, -1) = 3 > 0$

(3) $(2,\ 2.5)$ no, although the law holds (-2): at p^0 it costs $11.5 < e(p^0, u) = 12$

MWG Proposition 3.E.4 and (3.E.5), p. 62: u continuous, $\succsim$ locally non-satiated, $h(p, u)$ single-valued.

Slide 15 (optional) Both prices move: which bundles can be the new h?

No utility function is given, and none is needed: the test is the definition of h at each price vector. Because h^0 is the cheapest way to u at p^0, any bundle that reaches u costs at least $e(p^0, u)$ there; because h^0 still reaches u, the new cheapest bundle costs at most $p^1 \cdot h^0$ at p^1. Subtracting the two is slide 14's board proof, so the law $(p^1 - p^0) \cdot (h^1 - h^0) \leq 0$ is a consequence of the two cost checks, never a substitute for them. With both prices moving, reading "never rises in its own price" good by good says nothing: good 2 became cheaper at the same time.

That is why row 3 is the trap: it passes the law and still fails, because at p^0 it would undercut h^0 while reaching the same u. Row 1 passes both checks, and the checks are the whole story: each bundle lies above the other's cost line, which is exactly what lets a strictly convex curve pass through both with the right slopes, the chord between them being steeper than the tangent at h^0 and flatter than the one at the candidate. Row 2 fails the law outright, and both checks with it.

Break

Back at 11:00

slide 16 of 25

Slide 16 Break

Ten minutes.

Shephard's lemma and duality

MWG 3.F, 3.G

slide 17 of 25

Slide 17 Shephard's lemma and duality

Part 2: Shephard's lemma, the price picture behind it, compensated price effects, and the cost of living. MWG 3.F and 3.G.

Shephard's lemma: compensated demand is the price slope of e

$$h_\ell(p,u) = \frac{\partial e(p,u)}{\partial p_\ell}$$

u continuous; $\succsim$ locally non-satiated and strictly convex

$e = 2\sqrt{u\,p_1 p_2}$: $\dfrac{\partial e}{\partial p_1} = \sqrt{u\,p_2/p_1} = h_1$

slide 9

Roy (class 5): $x_\ell = -\dfrac{\partial v/\partial p_\ell}{\partial v/\partial w}$ Shephard: $h_\ell = \dfrac{\partial e}{\partial p_\ell}$

Roy divides to turn utils into euros; e is in euros already.

MWG Proposition 3.G.1 (Shephard's lemma), p. 68; Roy's identity: Prop. 3.G.4, p. 73.

Slide 18 Shephard's lemma: compensated demand is the price slope of e

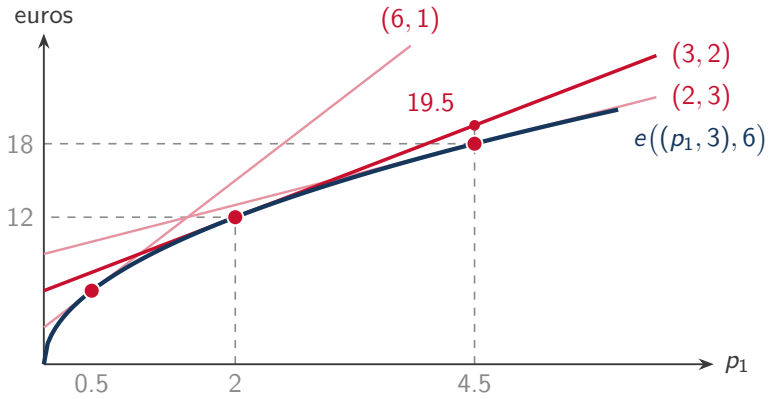
Read it as a sentence: the derivative of the least cost with respect to a price is the quantity of that good in the cheapest bundle. The check is slide 13's third column, which was h_1 from slide 9; on Exercise 1 the derivative of $u(p_1+p_2)$ in p_1 is u, the number of pairs she buys.

Why it holds: if p_1 rises by a cent, keeping the old bundle costs h_1 cents more, and switching to a better bundle saves nothing to first order, because at the cheapest bundle the cost line is tangent to the curve, the envelope argument of class 5's slide 14. The assumptions buy a unique cheapest bundle: at a kink of e, like row 2 of Exercise 1 at $p_1 = p_2$, there is no derivative and h is a set; local uniqueness is enough (MWG footnote 15: Leontief is not strictly convex, yet its h is unique).

Roy needed a ratio and a minus sign because v is in utils and the wealth slope converts them to euros; e is in euros already, so Shephard has neither. A minus sign, or a division by $\partial e/\partial u$, is the slip to avoid.

e is the lowest of straight lines: concave in p, slope $= h$

$u = x_1x_2$, $u = 6$, $p_2 = 3$: each straight line is the cost $p_1x_1 + 3x_2$ of one bundle on the curve



At $p_1 = 2$ the line of $(3,2)$ touches e with slope $3 = h_1$: Shephard's lemma, drawn.

MWG Figure 3.E.2, p. 60; Figure 3.F.3 and the duality theorem (Prop. 3.F.1), pp. 65–66.

Slide 19 e is the lowest of straight lines: concave in p, slope $= h$

Fix $u = 6$ and $p_2 = 3$ and let p_1 move. Each bundle on the curve, held fixed, costs $p_1x_1 + 3x_2$: a straight line in p_1 whose slope is that bundle's x_1. The expenditure function is the cheapest of these lines at each p_1, their lower envelope, and a minimum of straight lines is concave: that is Proposition 3.E.2's concavity, with no assumption on preferences at all. Where a bundle is the cheapest its line touches e, so the slope of e there is that bundle's x_1: Shephard's lemma is this picture.

The hook in one look: the red line is the cost of the old basket $(3,2)$ held fixed, and at $p_1 = 4.5$ it says 19.5 while the least cost of staying on $u = 6$ is 18. A fixed basket is a tangent line; the truth is the concave curve below it; the gap is the substitution bias. MWG's duality theorem (3.F): the set on or above the curve is the intersection of all half-planes $p \cdot x \geq e(p,u)$, one per price vector, so with convex preferences $e(\cdot,u)$ and the curve carry the same information. The trap: concavity of e comes from taking a minimum over bundles, not from any concavity of u.

Compensated price effects: own effects ≤ 0, cross effects equal

$$D_p h(p,u) = \begin{pmatrix} \partial h_1/\partial p_1 & \partial h_1/\partial p_2 \\ \partial h_2/\partial p_1 & \partial h_2/\partial p_2 \end{pmatrix} = \begin{pmatrix} -\frac{3}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{3} \end{pmatrix}$$

$x_1 x_2$ at $p = (2,3)$, $u = 6$

own effects ≤ 0

negative semidefinite

$\partial h_1/\partial p_2 = \partial h_2/\partial p_1$

symmetric

$D_p h\ p = 0$: $\quad -0.75 \cdot 2 + 0.5 \cdot 3 = 0$

degree 0 in p

Two goods: $\partial h_1/\partial p_2 = -(p_1/p_2)\,\partial h_1/\partial p_1 \geq 0$.

MWG Proposition 3.G.2, p. 69: $D_p h(p,u) = D_p^2 e(p,u)$.

Slide 20 Compensated price effects: own effects ≤ 0, cross effects equal

By Shephard's lemma the price derivatives of h are the second derivatives of e, and all three properties come from that. A concave e has a negative semidefinite matrix of second derivatives, so own effects are never positive, the compensated law of demand again. Second derivatives commute, so the two cross effects are equal, a symmetry nobody would guess from economics alone (MWG, after Samuelson 1947); here both are $\frac{1}{2}\sqrt{u/(p_1 p_2)} = \frac{1}{2}$. And h does not move when all prices scale, so each row times p is zero.

With two goods the last two facts force the takeaway: $p_1\,\partial h_1/\partial p_1 + p_2\,\partial h_1/\partial p_2 = 0$ with a nonpositive first term, so the cross effect is nonnegative; check: $-(2/3)(-\frac{3}{4}) = \frac{1}{2}$. Class 5's gross effects had no such symmetry. The same matrix built from observable Walrasian demand is next Tuesday's Slutsky matrix.

Net substitutes: with two goods, never net complements

net substitutes: $\partial h_1/\partial p_2 > 0$; net complements: < 0

CES: $u = \left(\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho\right)^{1/\rho}$, $\sigma = 1/(1-\rho)$; $\sigma = 2$: same $\gtrsim$ as $\sqrt{x_1} + \sqrt{x_2}$

σ	u	gross: x_1, p_2: $3 \to 4$	net: h_1, p_2: $3 \to 4$
	$p_1 = 2$; gross at $w = 12$ (class 5); net at the utility $w = 12$ bought		
0	$\min\{x_1, x_2\}$	$2.4 \to 2$: complements	$2.4 \to 2.4$: neither
1	$x_1 x_2$	$3 \to 3$: neither	$3 \to 3.46$: substitutes
2	$\sqrt{x_1} + \sqrt{x_2}$	$3.6 \to 4$: substitutes	$3.6 \to 4.44$: substitutes

Gross and net can disagree; the gap is a wealth effect: next Tuesday.

MWG 3.G, p. 70 (weak inequalities; strict here, so zero is "neither"). Class 5, slide 23; CES: class 3, slide 26.

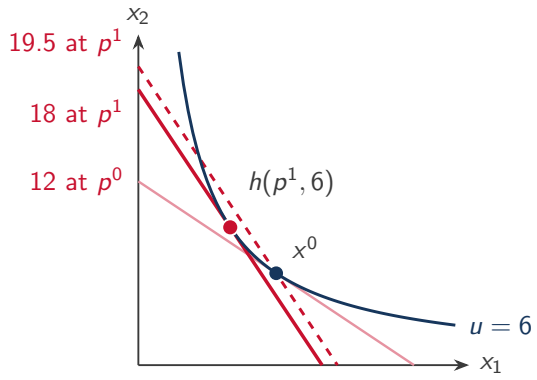
Slide 21 Net substitutes: with two goods, never net complements

Class 5 defined substitutes and complements with Walrasian demand and called them gross; MWG's own definition uses Hicksian demand, utility held fixed, and the two can disagree. The table repeats class 5's slide 23, now at the utility that $w = 12$ bought.

Leontief buys the goods one for one and at fixed utility keeps buying 2.4 of each, so her net cross effect is zero, neither in the slide's strict wording (MWG's weak inequalities would make it both); gross, the dearer pair made her poorer and she bought less of both. For $x_1 x_2$ nothing happens gross, since spending on good 1 is $w/2$ whatever p_2, but at fixed utility she slides along the curve towards good 1: net substitutes. Every net number sits above its gross number because both goods are normal: a dearer good 2 also makes her poorer, which pulls x_1 down, and that gap is next Tuesday's wealth effect.

With two goods the net column can never read complements, by slide 20, and net complements need at least three goods; MWG's "every good has at least one substitute" holds in its weak sense (≥ 0), while in the slide's strict wording Leontief has none, only neither.

True cost of living: $\times 1.5$; the fixed basket says $\times 1.625$



true index: $\dfrac{e(p^1, u^0)}{e(p^0, u^0)} = \dfrac{18}{12} = 1.5$

fixed basket (Laspeyres):

$$\frac{p^1 \cdot x^0}{p^0 \cdot x^0} = \frac{19.5}{12} = 1.625$$

x^0 still reaches u^0, so $e(p^1, u^0) \leq p^1 \cdot x^0$: never below the truth

To stay on her curve she needs 6 more euros, not 7.5.

Konüs (1924; Econometrica 1939): the true index. Own numbers: $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (4.5, 3)$.

In pairs: good 1 now costs 8: how much richer must each be?

$p^0 = (2, 3)$, $w = 12 \;\to\; p^1 = (8, 3)$. Fixed-basket index and true index.

$u(x)$	x^0	fixed basket $\dfrac{p^1 \cdot x^0}{12}$	true $\dfrac{e(p^1, u^0)}{12}$	
$\min\{x_1, x_2\}$	(2.4, 2.4)	2.2	2.2	no substitution, no bias
$x_1 x_2$	(3, 2)	2.5	2	$h(p^1, 6) = (1.5,\ 4)$
$x_1 + x_2$	(6, 0)	4	1.5	switch to good 2: (0, 6)

Seven minutes.

The more she can substitute, the more the fixed basket overstates.

Slide 22 True cost of living: $\times 1.5$; the fixed basket says $\times 1.625$

The cost of living for one consumer: how many euros she needs at the new prices to be exactly as well off as before, relative to what she spent, $e(p^1, u^0)/e(p^0, u^0)$, Konüs's true index. Before: $p^0 = (2, 3)$, 12 euros, bundle $(3, 2)$, $u^0 = 6$; then good 1 rises to 4.5. A fixed basket asks what $(3, 2)$ costs now, 19.5; but she need not keep $(3, 2)$, and the cheapest way back to $u = 6$ is $(2, 3)$ at 18.

The fixed basket never understates: the old basket still reaches u^0, so the least cost is at most its cost, and in the base period both equal w by slide 10. In the picture the dashed line through x^0 and the solid line touching the curve are parallel, and the gap of 1.5 euros is the substitution she is allowed; on slide 19 it was the gap between the tangent line and the curve at $p_1 = 4.5$.

An index that keeps an earlier basket sits on the line; one whose basket follows purchases comes closer to the curve, approximately, not exactly: slide 6's 0.2 points a year is this kind of gap, the substitution between spending categories. For $x_1 x_2$ the index does not depend on the base level u^0, since e is proportional to $\sqrt{u}$; in general it does.

Slide 23 In pairs: good 1 now costs 8: how much richer must each be?

Good 1 quadruples, from 2 to 8, for the three consumers of Exercise 1 and slide 9, and the question is how much richer each must be made: the old bundle priced at the new prices against the cheapest way back to the old utility level, each divided by the old spending of 12. The expenditure functions are already on the table.

Row 1: with no substitution possible the fixed basket is exact, 26.4 either way, index 2.2. Row 2: $(3, 2)$ now costs 30, index 2.5, but the cheapest way back to $u = 6$ is $(1.5, 4)$ at 24, index 2. Row 3: $(6, 0)$ now costs 48, index 4, yet she is indifferent between the goods unit for unit, so she moves entirely to good 2, $(0, 6)$ at 18, index 1.5.

Read the two columns against each other for $\sigma = 0, 1, \infty$: the easier the substitution, the larger the fixed basket's overstatement, and it is never negative. The slips: dividing by the new cost instead of the old; pricing the new bundle at the old prices and calling that the true index; keeping $(6, 0)$ in row 3 because she likes good 1.

(optional) Price her new basket: how does that index compare?

Slide 22's consumer: $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (4.5, 3)$.
At p^1 she buys x^1, with utility u^1.

(1) x^1, and the Paasche index $P = \dfrac{p^1 \cdot x^1}{p^0 \cdot x^1}$. $x^1 = (4/3,\ 2)$; $P = 18/13 \approx 1.38$

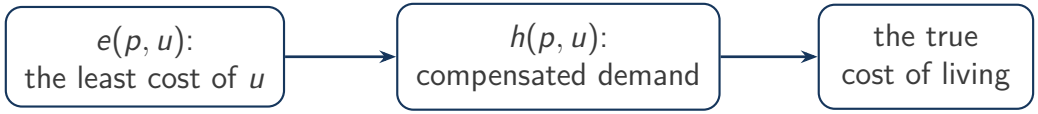
(2) Show, under the assumptions of Prop. 3.E.1: $P \leq \dfrac{e(p^1, u^1)}{e(p^0, u^1)}$. $e(p^1, u^1) = 12 = p^1 \cdot x^1$ (slide 10); x^1 reaches u^1, so $e(p^0, u^1) \leq p^0 \cdot x^1$

(3) Order P and slide 22's two indices. $P \approx 1.38 < 1.5$ (true) < 1.625 (fixed basket)

Today in one line

Shephard's lemma: $h_\ell = \partial e / \partial p_\ell$

$e(p^1, u^0)/e(p^0, u^0) \leq$ fixed basket



Tuesday: the Slutsky equation and the Slutsky matrix; summary of the relationships; compensating and equivalent variation.
Problem set 2: handed out in class today, one per group; due at the start of class, Wednesday 21 October, on paper.

Read MWG 3.G and 3.I (or Muñoz-Garcia ch. 2) for Tuesday 20 October.

Slide 24 (optional) Price her new basket: how does that index compare?

Slide 22's argument, run from the other end. There the old basket still reached u^0 at the new prices, so pricing it overstated the least cost of u^0; here the new basket still reaches u^1 at the old prices, so pricing it there overstates the least cost of u^1, and that overstatement sits in the denominator of P. The numerator is exact because at p^1 she spends her 12 euros as well as she can, slide 10's first identity, which is where local non-satiation enters. So a new-basket index never overstates the true index measured at the new utility, as an old-basket index never understates the one measured at the old utility.

For $x_1 x_2$ the true index is 1.5 whichever utility level is held fixed, since e is proportional to $\sqrt{u}$, so the two baskets bracket it from below and from above; in general the two bounds refer to two different utility levels, u^1 and u^0, and need not enclose a single number. The slips: dividing by the old spending of 12 instead of by what the new basket costs at the old prices, putting $e(p^0, u^0)$ in the denominator of the true index at u^1, and concluding that every fixed basket overstates.

Slide 25 Today in one line

In one line: the expenditure problem is the utility problem read backwards, fix a curve and find the cheapest bundle; its value $e(p, u)$ is in euros, degree 1 and concave in prices, and its solution $h(p, u)$ never rises in its own price; e and v are inverses; Shephard's lemma gets h from e with one derivative where Roy needed two; concavity of e is why a fixed basket overstates the cost of living; and with two goods the goods are never net complements: substitutes, or neither as for Leontief.

Tuesday 20 October: how h and x are tied together, the Slutsky equation and the Slutsky matrix; the full map of x, v, h, e and the arrows between them; and money measures of a price change, compensating and equivalent variation. Reading: MWG 3.G and 3.I, or Muñoz-Garcia chapter 2.

Problem set 2 is handed out on paper at the end of class, one per group; it is due at the start of class on Wednesday 21 October, on paper, one copy per group.